



-1-

14. Strain gauge has a _____
 A) low temperature coefficient of resistance
 B) high temperature coefficient of resistance
 C) zero temperature coefficient of resistance
 D) infinite temperature coefficient of resistance
15. Temperature transducers make use of _____
 A) change in resistivity
 B) change in length
 C) change in area
 D) change in capacitance
16. Load cell is used for the measurement of _____
 A) Area
 B) force
 C) mass
 D) length
17. The relation between mating parts is called _____
 A) Connection
 B) Fits
 C) Joints
 D) Link
18. Upper deviation is the algebraic difference between the _____ and corresponding basic size.
 A) actual size
 B) minimum limit of the size
 C) maximum limit of the size
 D) design size
19. Lower deviation is the algebraic difference between the _____ and corresponding basic size.
 A) actual size
 B) minimum limit of the size
 C) maximum limit of the size
 D) design size
20. The difference between the maximum and minimum permissible limits of the sizes is called _____
 A) Deviation
 B) Allowance
 C) Tolerance
 D) Actual deviation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Outline the objectives of metrology.	4	L*2	1	1
	b) Briefly explain the subdivision of standards.	4	L2	1	1
	c) Explain the terminology of screw thread with neat sketch.	8	L2	2	2
2.	a) Briefly explain the wavelength standard.	4	L2	1	2
	b) Obtain M87 set of slip gauges and build the dimension of 56.3835.	4	L1	1	2
	c) Derive an expression for effective diameter of screw threads by 3-wire methods.	8	L3	2	2
3.	a) Give the combinations of gauges required to build: 57° 34' 9".	4	L1	1	2
	b) Sketch and explain the principle of optical flats.	4	L2	1	2
	c) Explain the working principle of Zeiss ultra optimeter and highlight its magnification.	8	L2	2	1
Unit – II					
4.	a) Enumerate the generalized measurement system with suitable example.	8	L2	3	2
	b) With a neat sketch explain the working principle of analytical balance.	8	L1	4	2
5.	a) Explain the working principle of CRO with neat sketch.	8	L2	3	1
	b) Describe the McLeod gauge with a neat sketch.	8	L2	4	2
6.	a) Explain the telemetry transmitting and receiving system.	8	L2	3	1
	b) Sketch and explain the working principle of optical Pyrometer.	8	L1	4	2

Unit – III

7. a) Calculate all the relevant dimensions of 35H7/f8 fit,
- Dimension 35 mm falls in the step of 30-50 mm.
 - The fundamental deviation for f shaft is $-5.5D^{0.41}$.
 - i (in microns) $=0.45(D)^{1/3}+0.001D$,
 - $IT_7=16i$ and $IT_8=25i$.
- | | | | | |
|---|----|----|---|---|
| | 10 | L3 | 5 | 2 |
| b) With necessary sketches explain types of fits and their designation. | 6 | L2 | 5 | 1 |
8. a) Determine the dimension of the shaft and hole for a fit 30H₈/d₁₀ and sketch the fit, given the following data:
- Diameter 30 falls in the diameter range 18-30mm, Upper deviation for 'd' shaft is $-16D^{0.44}$
 - $i = 0.45(D)^{1/3}+0.001D$, $IT_8 = 25i$ and $IT_{10} = 64i$.
- | | | | | |
|--|----|----|---|---|
| | 10 | L3 | 5 | 2 |
| b) Discuss the principle of interchangeability and selective assembly. | 6 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
